LORA

(low rank adaptation)

two types of training

1) pre-training
2) Post-training

Lora is a Post-training Process in which we fine-tune Large Language Model for Several Application

eg:- 1) Custom chat model model
2) Internal Company model

it is process of fine tuning a model for Our domain

why we need Lora ?

Just think of training a 5 Billion parameters model

Just think of training a 5 Billion parameters model it will cost many Computational power and Storage

to store a 5 billion parameter you need 20 GB - 30 GB Storage and a good GPU

Heres is the part

Suppose there a model trained on English, Spanish We want to learn/train the model with our Mother tongh and we didn't have Enogh Computational Power / GPU's ,How Can we Solve this issue

Here Comes the Significance of Lora / Lora as parameter effecient fine tuning which means we are training one new data on parameter - effecient way

How?

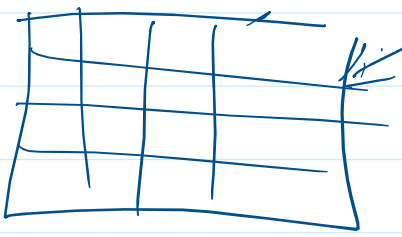
when I get into the Programming I really what to know how it is working So I read many books, watched many videos

Thanks to Sebastian raschka book on LLM & finaly understand how it works

Here The interesting Part

Suppose this is One layer

LLM — 5 Billions

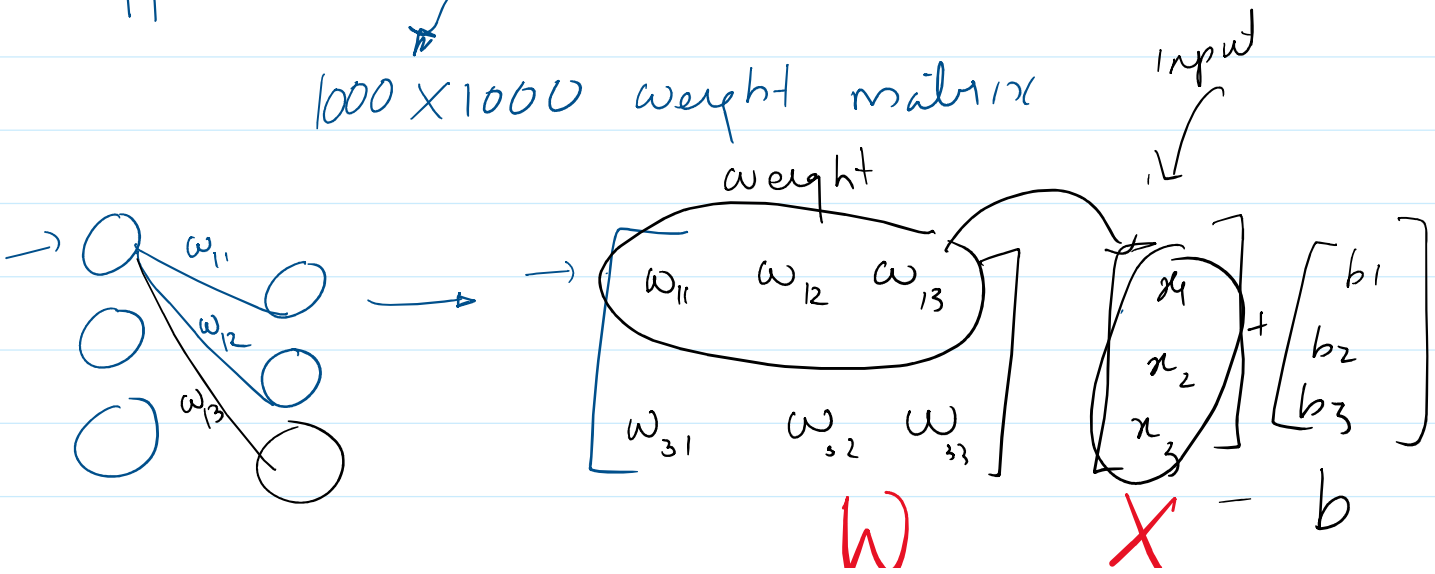


W matrix

5 Billion parameters converted 1000 × 1000

Suppose there is a

1000 × 1000 weight matrix



W · X − b

In normal training process, the matrix multiplication is Calculate parallely (thanks to Cuda)

So $Y = [W \cdot X] + b$

One bias is broadcasted across the (weight matrix · X)

activation functions are also used

So when we trained Our 5 billion P LLM it cost huge Comput

1000 × 1000 = 1000000 operations

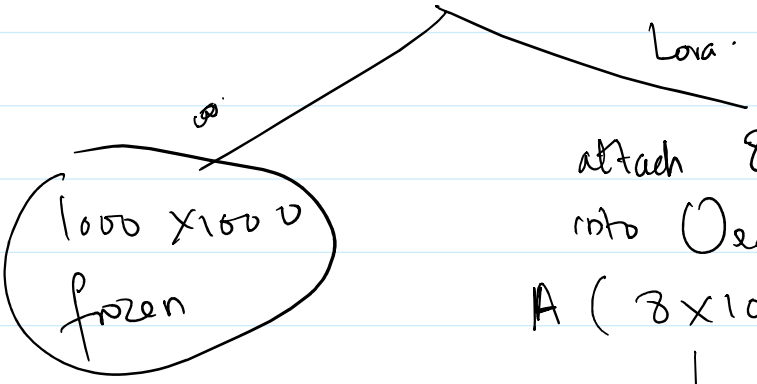
what is Lora doing

We make the Orginal weight matrix into Frozen (means it is not trained)

Process

in Suppose 1000 × 1000 is our Orginal matrix

In Suppose 1000 × 1000 ... our Original matrix



attach 8 row matrix into Original matrix (A matrix)

A (8 × 1000) Suppress the Input

↓

A matrix is multiplied by (1000 × 8) B

Ax.

x = (1, 1000)

A = (8 × 1000)^T → (1000 · 8)

x A^T = 1 × 1000 ⊗ 1000 · 8

[1 × 8] intermediate

B = (1000 × 8) | we need the Exact shape of output

1 × 1000 ← 1 · 8 × 8 × 1000

m · m

(exact shape of output

So 1000 × 1000 → 1000000 + ~~8 · 1000 · 1000 · 8~~ 16,000

1,016,000 Parameters

OR

or

16,000 trainable parameters

In Back propagation only the trainable parameters are updates

Final Equation

$$y_{final} = Wx + \frac{\alpha}{r} x A^T B^T$$

r = rank ← 8 in this case

α : Scaling parameter

Eg if rank is 16
α is 32 $= 2$

$$\text{Eq} \rightarrow y_{fn} = Wx + 2BAx$$

Thank You

